

Fig. 3: An actual-size, single-side combined PCB for transmitter and receiver circuits

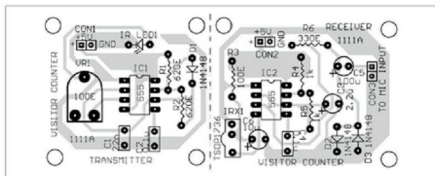


Fig. 4: Component layout for the PCB

on the screen. It generates the values in text format in command window. If the values obtained are not 4.05 and 172.0, check power supply to the transmitter

and receiver circuits (it should be 5V) and also vary VR1 in the transmitter circuit to get proper threshold values. Check whether there is any interfacing

problem with your sound card.

The threshold values have to be used in the main program (personcounter.m) as follows:

```
Fs=8000;
recordtime=1000;
lower=4.05;
upper=172;
PLAYSOUND=0;
```

Once you have done the settings as given above, run 'personcounter.m' and switch on the circuits to start counting. 'personcounter.m' is a GUI-based program that displays the number of visitors on the computer screen.

An actual-size, single-side combined PCB for transmitter and receiver circuits is shown in Fig. 3 and its component layout in Fig. 4.

The PCB for the transmitter circuit can be separated by cutting along the dotted line shown in Figs 3 and 4.

EFY note. 1. Don't forget to use diodes D2 and D3 in the circuit. Otherwise, your sound card may malfunction. It may even cause permanent damage to the sound card.

2. The source codes of this article are available on www.efymag.com website.

PERSONCOUNTER.M

```
function personcounter
Fs=8000;
recordtime=1000;
lower=4.05;
upper=172.00;
PLAYSOUND=0;
t=1:2000;
sound = 2*sin(2*pi*t*1000/Fs);
count=0; % when you restart the program,
reload the old value here
sel=0;
STOPPED = 1;

% Figure Window
hfig = figure('Name','Person Counter',...
'NumberTitle','off',...
'Position',[600 500 350 100],...
'Menubar','none',...
'Resize','off',...
'KeyPressFcn','@keyPressFcn',...
'CloseRequestFcn','@closeRequestFcn');

% Menus
filemenu = uimenu('Label','File');
uihandles(filemenu,'Label','About','Callba
ok','@aboutFcn');
uihandles(filemenu,'Label','Exit','Callback',@
closeRequestFcn);

% Buttons
uicontrol(hfig,'Style','PushButton',...
'Position',[110 10 75 25],...
'String','START',...
'Callback','@startFcn');
uicontrol(hfig,'Style','PushButton',...
'Position',[140 10 75 25],...
'String','RESET (R)',...
'Callback','@resetFcn');

uicontrol(hfig,'Style','PushButton',...
'Position',[125 10 75 25],...
'String','EXIT (X)',...
'Callback','@closeRequestFcn');

% Stopwatch Time Display
DISPLAY = uicontrol(hfig,'Style','text',...
'Position',[10 45 330 55],...
'BackgroundColor',[0.8 0.8 0.8],...
'FontSize',35);

set(hfig,'HandleVisibility','off');

str = num2str(count);
set(DISPLAY,'String',str);

% Start the Timer
htimer = timer('TimerFcn',@
timerFcn,'Period',@recordtime/5000,'En
ExecutionMode','FixedRate');
start(htimer);

function timerFcn(varargin)
if ~STOPPED
y=waverecord(1000,8000);
mean(y(50:end,1));
power=sum(y.^2);
if (power<lower)
sel=1;
else if (power > upper &&
sel==1)
count=count+1;
sel=0;
str = num2str(count);
set(DISPLAY,'String',str);
if (PLAYSOUND)
wavplay(sound,Fs);
end;
end;
end;

function startFcn(varargin)
% Start the Stopwatch
STOPPED = 0;
end

function resetFcn(varargin)
% Reset the Stopwatch
STOPPED = 1;
count = 0;
str = num2str(count);
set(DISPLAY,'String',str);
end

function aboutFcn(varargin)
msgbox('Person Counter Using
MATLAB GUI Version','...
by Karthikyan N <karthi.
amrita.cbe@gmail.com>','About');
end

function closeRequestFcn(varargin)
% Stop the Timer
try
stop(htimer);
delete(htimer);
catch errmg
rethrow(errmg);
end
% Close the Figure Window
closeReq;
end
end
```